

# SMRUTISHREE BEHERA

## PhD Researcher in Physics | Ferroelectric Materials & Neuromorphic Devices

SRM University-AP, India

+91 9861623218 | smrutishreebehera2001@gmail.com

smrutishreebehera.github.io

### RESEARCH PROFILE

Physics researcher pursuing a PhD at SRM University-AP, India, working on ferroelectric materials and their potential for neuromorphic devices and memory applications. Academic training spans condensed matter physics, solid-state physics and quantum mechanics, supported by experience in experimental characterization, electronics and scientific software. Developing interests include ferroelectric-enhanced neurotechnology, neural interfaces and neural modulation.

### RESEARCH INTERESTS

Ferroelectric materials and functional properties; material-based memory and neuromorphic devices; ferroelectric-enhanced neurotechnology and neural modulation.

### RESEARCH EXPERIENCE

#### Doctoral Research - Ferroelectric Materials

SRM University-AP, India | Ongoing

- Investigating the physical and functional properties of ferroelectric materials, with a focus on applications in neuromorphic hardware and memory devices.
- Developing research interests at the intersection of functional ferroelectric materials, neurotechnology and neural modulation.

#### Quark Model and Classification of Hadrons

Master's project | Kendrapara Autonomous College, affiliated to Utkal University | 3 months

- Explored SU(3) flavor symmetry and the classification of baryon and meson families.
- Compared theoretical predictions of hadron properties with experimental data, strengthening a foundation in quantum chromodynamics.

#### Design and Optimization of a Colpitts Oscillator Circuit

B.Sc. project | Berhampur University | 1 month

- Designed and studied oscillator circuits using transistor and op-amp setups to examine frequency stability and waveform generation.
- Investigated the effects of passive components on oscillation frequency and circuit performance, developing skills in analog electronics, circuit testing and signal analysis.

# SMRUTISHREE BEHERA

Academic background and technical skills

## EDUCATION

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### PhD in Physics

SRM University-AP, India | Ongoing

- Research focus: ferroelectric materials and functional properties for neuromorphic device applications. Developing interests: ferroelectric-enhanced neurotechnology and neural modulation.

### M.Sc. in Physics

Kendrapara Autonomous College, affiliated to Utkal University | Oct 2022 - Jul 2024

- CGPA: 7.35 (Grade B).
- Coursework: Advanced Condensed Matter Physics I & II, Basic Condensed Matter Physics, Advanced Quantum Mechanics, Statistical Mechanics and Dissertation/Project.

### B.Sc. - Physics Honours

Berhampur University | Jul 2019 - Jul 2022

- CGPA: 8.66 (Grade A+).
- Coursework: Solid State Physics, Quantum Mechanics & Applications, Nano Materials & Applications, Statistical Mechanics, Electromagnetic Theory, Elements of Modern Physics and Classical Dynamics.

### Higher Secondary School

Council of Higher Secondary Education, Odisha | 2018

- Result: 76.5%. Subjects: Physics, Chemistry, Mathematics and Biology (Botany & Zoology).

### High School

Board of Secondary Education, Odisha | 2016

- Result: 90.0%. Subjects: Mathematics, General Science, Social Science, English and Odia.

## TECHNICAL SKILLS

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**Electrical characterization:** Impedance analyzer; source meter.

**Structural and spectroscopic characterization:** X-ray diffraction (XRD); FTIR spectroscopy; Raman spectroscopy.

**Scientific software and instrumentation:** Origin; LabVIEW.

**Programming:** C; Fortran.

**Electronics:** Analog circuit design; circuit testing; signal analysis.

## ADDITIONAL INFORMATION

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**Languages:** English, Hindi and Odia.

**Paper review:** Which came first, "Galaxy or Blackhole".

**Interests:** Dancing, drawing, reading and listening to music.